# Comparing Mutual Information in Finite Samples

*Can MI-specific degrees of freedom improve comparisons between populations?*

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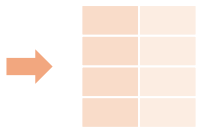
# Motivation: comparing MI across populations

Mutual information measures how much observing one variable reduces uncertainty about another. **Different sample estimates can arise from sampling noise.**

**Null hypothesis** $H_0 : I(P) = I(Q)$

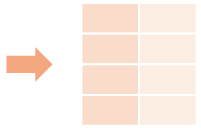
## Online shoppers

**P: new visitors**
1,694 sessions



corrected MI estimate
**0.0043 nats**

**Q: returning visitors**
10,551 sessions



corrected MI estimate
**0.0179 nats**

Variables: month of visit and purchase outcome.

## Statistical task

**Draw a count table from each population**

**Estimate MI from each table**

**Assess whether sampling noise can explain the difference**

## Does month–purchase MI differ between the populations?

*Month × purchase tables (10×2); corrected MI estimates in nats. Data: Sakar and Kastro (2018). Bossomaier et al. (2016)*

# Literature review: foundations and the MI-specific step

Existing results provide MI estimates, a normal-reference comparison and moment matching. **This project derives the MI variance sensitivity needed for its degrees of freedom.**

| | *MI estimation and uncertainty* | *Normal Wald comparison* | *Welch–Satterthwaite moment matching* |
|---|---|---|---|
| **Idea** | Estimate MI and first-order sampling variance | Divide the estimated difference by its standard error | Match variance moments to choose degrees of freedom |
| **Key work** | Moddemeijer (1989); Paninski (2003) | Mora and Ruiz-Castillo (2009) | Satterthwaite (1946); Welch (1947); Hutcheson (1970); Pan and Wall (2002) |
| **Required step** | Track variation in the variance estimate itself | Assess the fixed normal cutoff in finite samples | Calculate the MI-specific variance sensitivity |

**Contribution:** Derive that sensitivity and evaluate the resulting degrees of freedom.

*Thesis Sections 2.2–2.5 and 3.2–3.3; full references in B13–B14*

# Method: one statistic, two reference distributions

Both methods use **the same estimated MI difference and standard error**. Expanded Welch uses estimated degrees of freedom for a Student t reference.

### Shared statistic

$$T = \frac{\widehat{\Delta}_{\mathrm{BC}}}{\widehat{\mathrm{SE}}}, \qquad \widehat{\mathrm{SE}}^2 = \frac{\widehat{V}(P)}{n_P} + \frac{\widehat{V}(Q)}{n_Q}$$

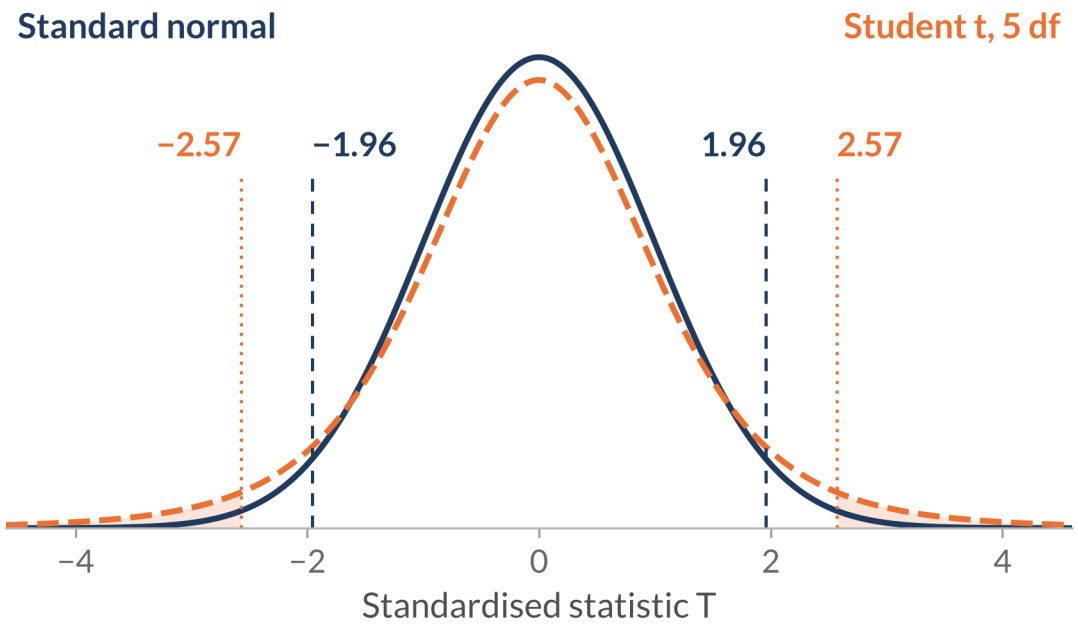
$\widehat{\Delta}_{\mathrm{BC}}$: estimated $I(P) - I(Q)$, after leading bias correction.

### Normal Wald

Standard normal distribution; two-sided 5% cutoff about **±1.96**.

### Expanded Welch

Student t distribution with estimated degrees of freedom; **a larger cutoff**.




*Student t distribution with 5 degrees of freedom, for illustration; cutoff about ±2.57.*

# Approach and contribution

Tracking how the pointwise-MI variance changes with the table gives the sampling-variance input for **MI-specific degrees of freedom**.

## Derivation chain

**Cell probabilities change**
Move a small amount of probability towards one cell.

→

**Margins and pointwise MI change**
Track their joint effect.

→

**Variance sensitivity, derived here**
$g_P(x, y)$ describes the change in $V(P)$.

→

**Degrees of freedom**
Use the spread of those sensitivities.

## Result and approximations

$$\mathrm{Var}\{\widehat{V}(P)\} \approx \frac{\tau^2(P)}{n_P} \quad \Longrightarrow \quad \hat{\nu}_P = \frac{2n_P\widehat{V}(P)^2}{\hat{\tau}^2(P)}$$

$\tau^2(P)$ is the variance of $g_P(X, Y)$. Repeat for Q and combine using Welch–Satterthwaite.

### Two approximations evaluated

1. First-order sampling expansion
2. Scaled chi-squared moment matching

*Thesis Sections 4.1–4.3; exact derivative in B1, moment matching in B2*

# Evaluation design

Known populations separate approximation accuracy, false positives and power. Both tests receive **the same sampled table pairs**.

## Research questions

1. **RQ1: Approximation accuracy**
   Sampling moments of $\hat{V}$ and component df, including near independence
2. **RQ2: False positives**
   Equal MI at 0.005, 0.02 and 0.30 nats
3. **RQ3: Power**
   P MI 0.02; MI-gap magnitudes 0.005, 0.01 and 0.02 nats

## Shared simulation grid

- Four sizes: 2×2, 4×4, 8×8, 16×16
- Uniform, same-skew and different-skew margins
- Ten sample sizes: 10 to 50,000 observations per population
- 100,000 independent table pairs per setting

**Model:** copying construction: choose margins and solve for target MI.

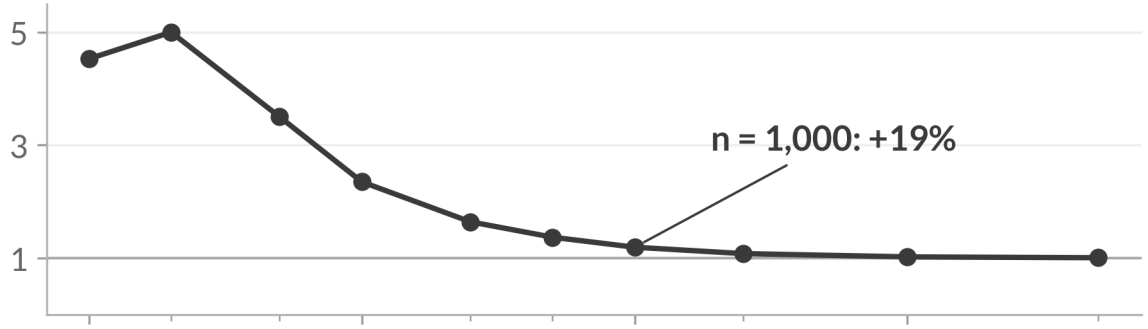
**Reporting:** usual 0.05 p-value thresholds. Invalid outputs count as non-rejections; validity is reported separately.

*Morris et al. (2019). Thesis Chapter 5, Tables 5.1–5.3; exact vectors, sample sizes and rules in B3*

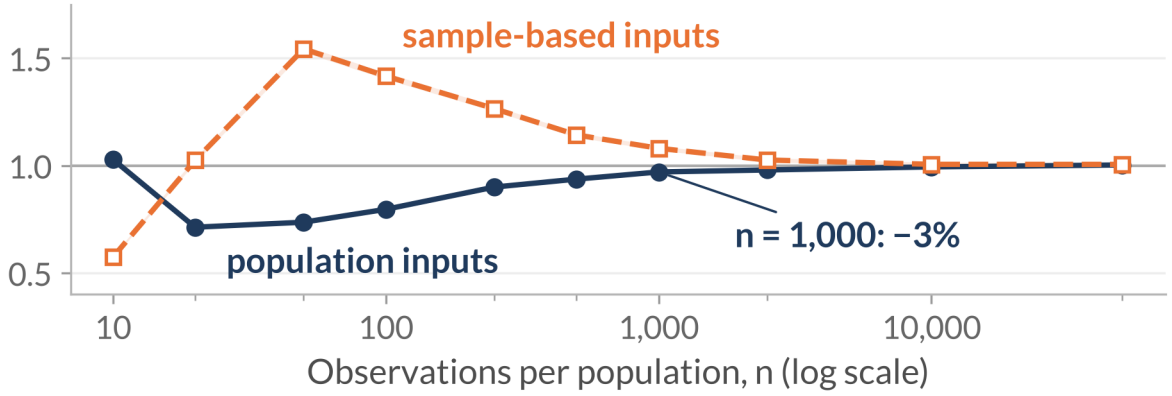
# Results: approximation accuracy

The mean of $\hat{V}$ can **remain inflated** even when its population first-order sampling-variance prediction is accurate.

**Mean of $\hat{V}$ / population $V$**




**Predicted / observed sampling variance of $\hat{V}$**




**Different-skew P · 4×4 · population MI 0.02 nats**

**Mean: rare cells can inflate it.**

Newly observed rare cells can have large pointwise-MI deviations, whose squares raise $\hat{V}$.

**Sampling variance: linearisation can miss large changes.**

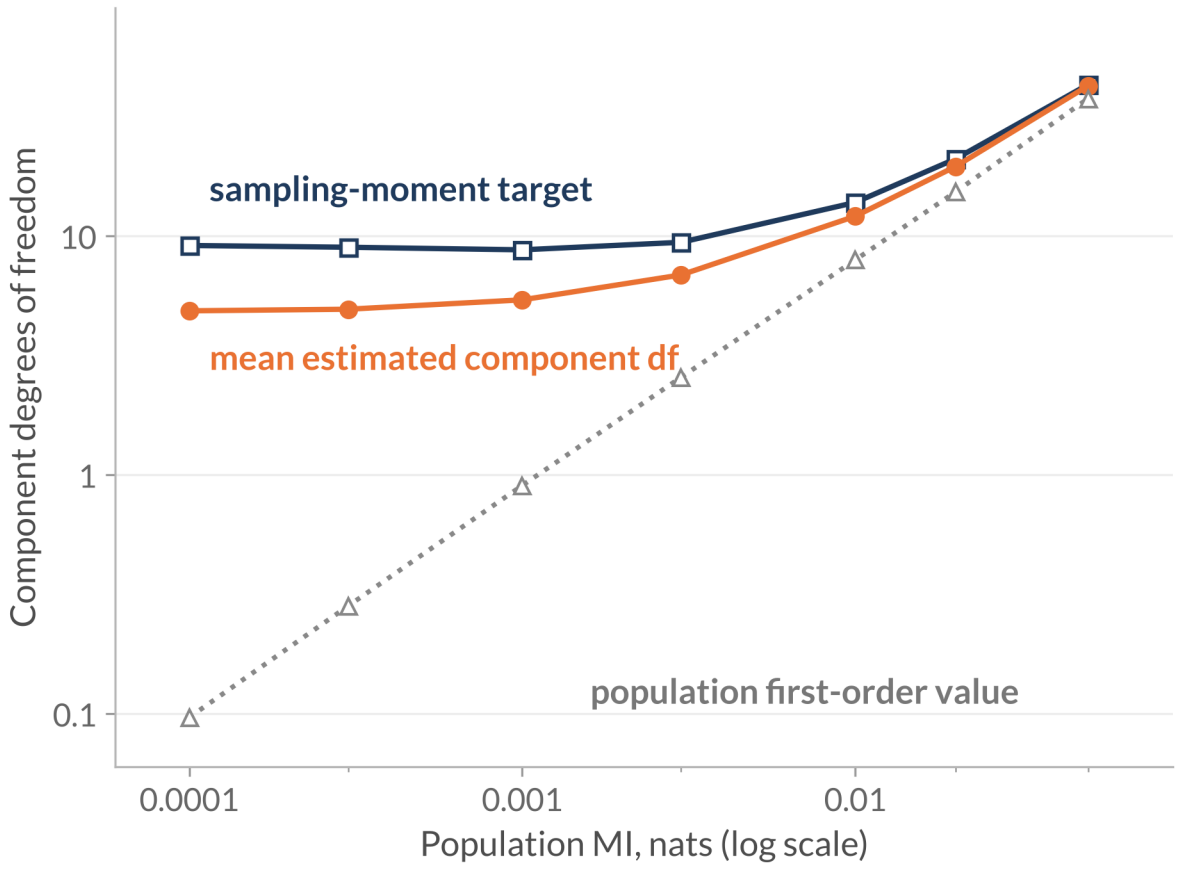
Rare cells appear in some samples and not others; sample-based sensitivities can also overpredict the resulting variation.

**At n = 1,000:** mean **19% high**; population variance prediction **3% low**. Both moments enter the degrees of freedom.

*Thesis Sections 6.1.1–6.1.2, Figures 6.1–6.2. Bands: approximate 95% Monte Carlo intervals. Other sizes and Q: B4*

# Results: near independence

Near independence, **sampling noise keeps the mean estimated component degrees of freedom positive** while the population first-order value becomes very small.




Different-skew P · 4×4 · n = 1,000 fixed

### Population first-order value

Insert known population inputs into the component formula. Near independence, its local value is approximately n × population MI.

### Sample estimate

Quadratic sampling errors create apparent association, which remains in the sample-based calculation.

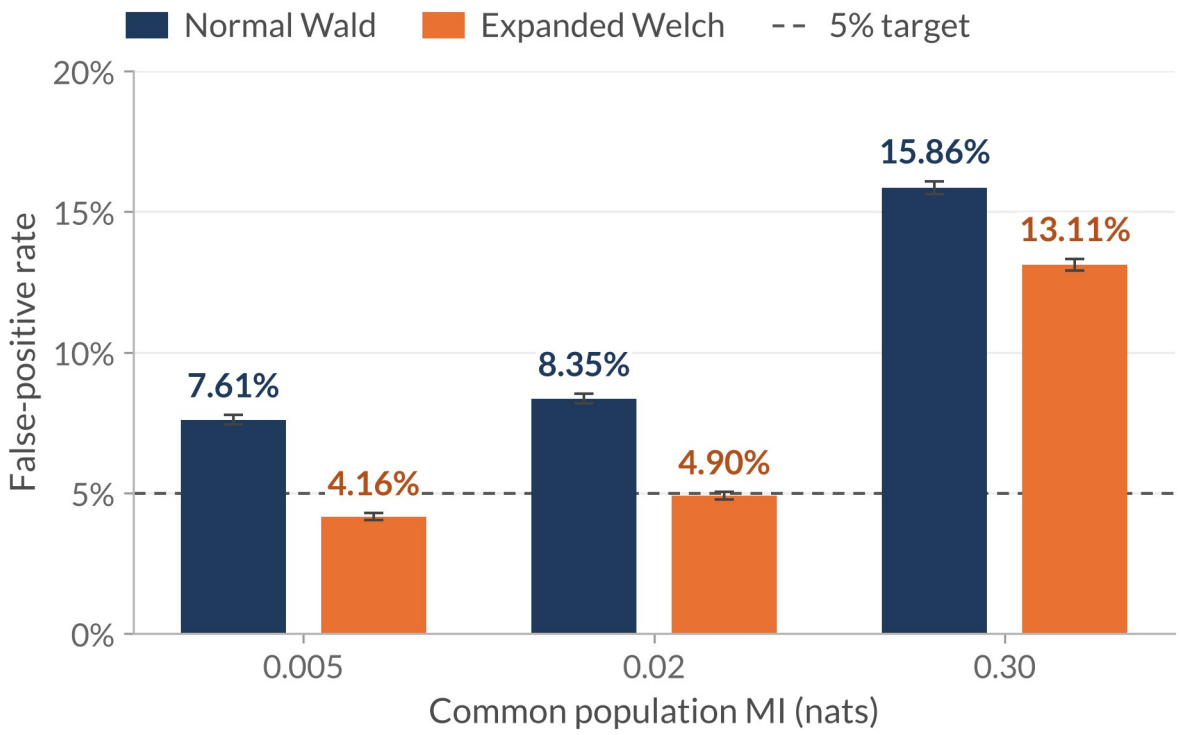
**At MI 0.0001:** mean estimated component df **≈4.9**; sampling-moment target **≈9.1**; population first-order value **≈0.1**

*Thesis Section 6.1.3, Figure 6.3; Appendix A.2. Bands: approximate 95% Monte Carlo intervals. Local calculation: B5*

# Results: false positives across common MI values

A larger cutoff **brings some excessive false-positive rates closer to 5%**, but accuracy at one association level does not persist across common MI values.

**Uniform 4×4 · n = 20 per population · equal MI in P and Q**




**Nominal target: 5%.** Both tests are valid in every repetition in these examples.

### Common MI 0.02: closer to 5%

Expanded Welch lowers the rate from 8.35% to 4.90%.

### Common MI 0.30: still above 5%

Expanded Welch still rejects in 13.11% of repetitions.

**Interpretation:** The larger cutoff reduces rejection, while the shared statistic can still have the wrong sampling distribution.

*Slices of the balanced null grid. Whiskers: pointwise 95% Monte Carlo intervals. Thesis Section 6.2.1, Figure 6.4; full grids in B6–B8*

# Results: why bias changes rejection

Similar errors in estimated uncertainty accompany **opposite false-positive behaviour** when bias in the MI difference changes.

**16×16 · n = 1,000 per population · equal MI 0.02 nats**

| Margins | Variance ratio | Bias / SD | Normal Wald | Expanded Welch |
|---|---|---|---|---|
| Same skew | 1.93 | ≈0 | 0.60% | 0.53% |
| Different skew | 1.94 | **2.06** | 26.01% | 24.84% |

*Variance ratio: average squared estimated standard error / observed variance of the estimated MI difference.*

### Same skew

The oversized denominator suppresses the statistic; expected MI biases cancel.

### Different skew

The estimated difference is shifted away from zero by unequal residual biases.

**Conclusion:** Changing the cutoff leaves the biased estimated difference unchanged. Both tests are valid in every repetition here.

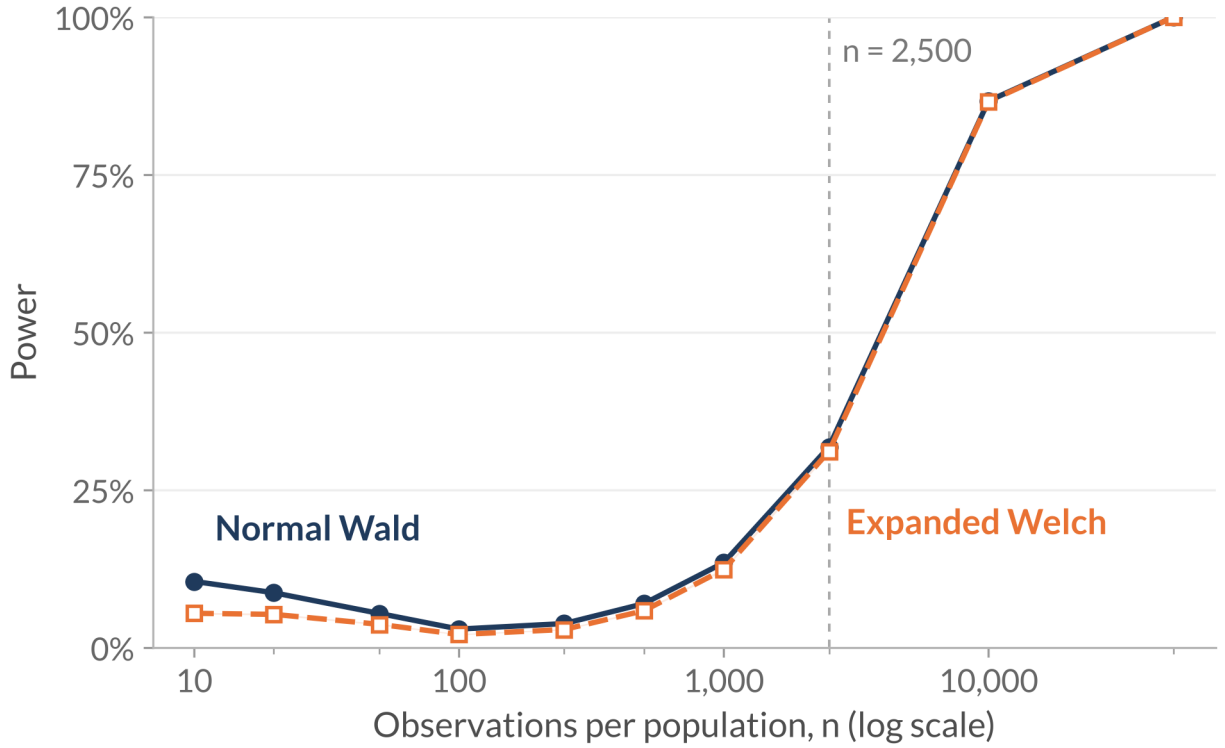
*Thesis Section 6.2.4, Table 6.1; current null rates and diagnostics. Full different-skew grid: B8*

# Results: power across MI differences

With sufficient observations, **larger MI differences become easier to detect**. Expanded Welch's larger cutoff lowers or preserves rejection.

**Uniform 4×4 · population MI in P = 0.02 nats**

**MI-gap magnitude 0.01 nats: power against sample size**




**Snapshot at n = 2,500**

| MI-gap magnitude (nats) | Normal Wald | Expanded Welch |
|---|---|---|
| 0.005 | 11.60% | 11.14% |
| 0.01 | 31.85% | 31.14% |
| 0.02 | 79.24% | 78.74% |

**Power:** probability of rejecting equal MI when population MI differs. Rates use the usual 0.05 p-value threshold.

*Rates divide valid rejections by all 100,000 repetitions; invalid outputs count as non-rejections.*

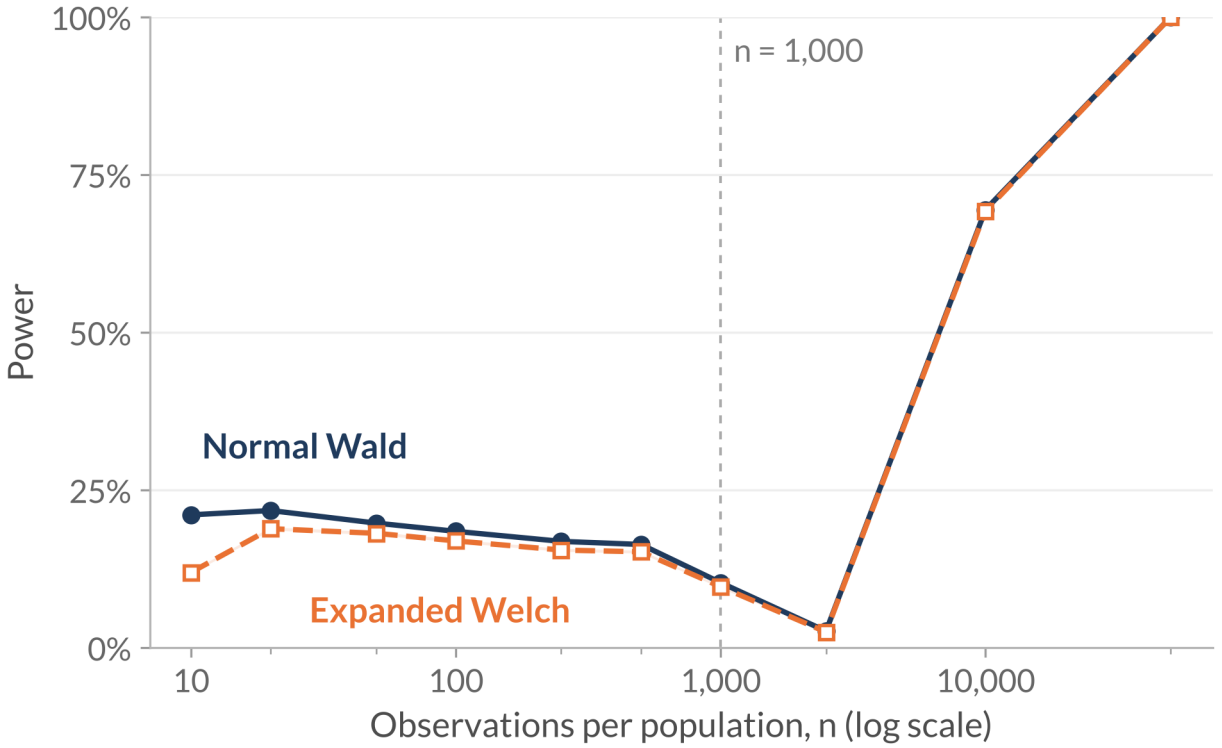
*Bands: pointwise 95% Monte Carlo intervals. Thesis Section 6.3.1, Figure 6.7; full grids in B9–B11*

# Results: bias can reverse the estimated direction

In sparse tables, unequal residual biases can outweigh the true MI difference and **reverse the estimated direction**.

**Different-skew 16×16 · MI 0.02 versus 0.03 nats**

**MI-gap magnitude 0.01 nats: power against sample size**




**At n = 1,000**

| | |
|---|---|
| True difference, P minus Q | **−0.0100 nats** |
| Mean estimated difference | **+0.0207 nats** |

Expanded Welch rejects in **9.66%** of repetitions.

**Among those rejections**, over 99.9% have the opposite estimated direction.

**Mechanism:** As positive bias diminishes, the sampling centre passes near zero, so rejection initially falls before recovering.

*At n = 1,000: 9,659 of 9,662 Expanded Welch rejections are positive; both methods 100% valid. Thesis Figure 6.9; full grid in B11*

# Illustration: online shopping data

Returning visitors have higher estimated month–purchase MI, and **both approximate tests reject equal MI after adjustment**.

**Month × purchase: corrected MI estimates (nats)**

New visitors

**0.0043**

Returning visitors

**0.0179**

Holm-adjusted p-values across six comparisons

**Normal Wald** 0.000189

**Expanded Welch** 0.00149

**Both tests reject equal MI after Holm adjustment**

**Weekend × purchase**

**Estimates are close; neither test rejects equality.**

**Interpretation:** Population MI is unknown. The simulations provide the evidence about reliability.

*Sakar and Kastro (2018): 1,694 new and 10,551 returning sessions. Holm (1979). Thesis Section 6.4; details in B12*

# Conclusions and next steps

MI-specific degrees of freedom improve some false-positive rates, while remaining failures identify **estimation accuracy as a priority**.

## Contribution and findings

1. Derived variance sensitivity and MI-specific degrees of freedom.
2. Brought some excessive false-positive rates closer to 5%; deepened conservatism elsewhere.
3. Reduced nominal power; residual bias can reverse the estimated direction.

## Next steps

1. Improve bias and uncertainty estimates, including quadratic variation near independence.
2. Test other association constructions and unequal sample sizes.

Thank you.

# Appendix

# Variance sensitivity

**Perturbation towards one cell:** $P_\varepsilon = (1-\varepsilon)P + \varepsilon\delta_{(x,y)}$

**Changing probability weights**

$$g_P(x,y) = \{\ell_P(x,y) - I(P)\}^2 - V(P)$$

**Changing pointwise-MI values and margins**

$$+ 2[\ell_P(x,y) - \mathbb{E}_P\{\ell_P(X,Y) \mid X = x\}$$

$$- \mathbb{E}_P\{\ell_P(X,Y) \mid Y = y\} + I(P)]$$

Pointwise MI: $\ell_P(i,j) = \log\{p_{ij}/(p_{i+}p_{+j})\}$

Sensitivity variance: $\tau^2(P) = \mathrm{Var}_P\{g_P(X,Y)\}$

*Exact for positive population tables; its use for sampling errors is first order. Thesis Chapter 4*

# Moment matching and combined degrees of freedom

## Component degrees of freedom

$$\nu_{P,\,\mathrm{mom}} = \frac{2[\mathbb{E}\{\widehat{V}(P)\}]^2}{\mathrm{Var}\{\widehat{V}(P)\}}, \qquad \widehat{\nu}_P = \frac{2n_P\widehat{V}(P)^2}{\widehat{\tau}^2(P)}$$

## Combined degrees of freedom

$$\widehat{\nu} = \frac{\{\widehat{V}(P)/n_P + \widehat{V}(Q)/n_Q\}^2}{\{\widehat{V}(P)/n_P\}^2/\widehat{\nu}_P + \{\widehat{V}(Q)/n_Q\}^2/\widehat{\nu}_Q}$$

Moment matching does not establish the normality or independence needed for an exact Student t distribution.

*Satterthwaite (1946). Thesis Sections 4.1 and 4.3*

# Population construction, grid and validity

$$p_{ij}(\lambda) = (1-\lambda)a_i a_j + \lambda a_i \mathbf{1}\{i=j\}, \qquad 0 \le \lambda < 1$$

## Parameter grid

- k = 2, 4, 8, 16; uniform, same-skew or different-skew margins
- Same skew: both use $m_k(0.8)$. Different skew: P uses $m_k(0.7)$, Q uses $m_k(0.8)$, where $m_k(\rho) = (\rho, (1-\rho)/(k-1), \ldots)$
- Null: common MI 0.005, 0.02, 0.30. Power: P MI 0.02, Q MI 0.025, 0.03, 0.04

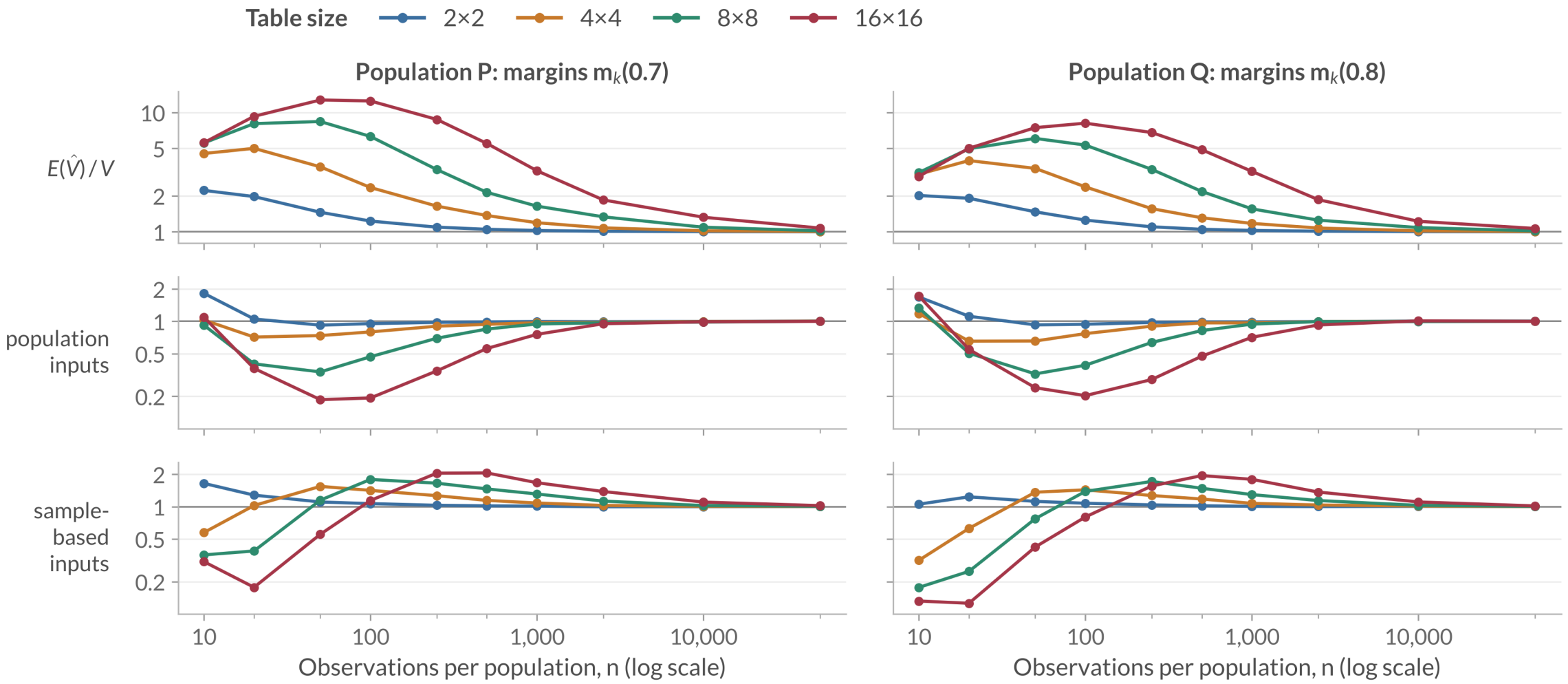
## Validity and rates

- Each study has 360 settings, 100,000 independent pairs per setting
- Wald needs a finite difference and positive finite standard error. Expanded Welch also needs valid positive finite df calculations.
- Rate = valid rejections / all repetitions. Report validity separately.

**n =** 10, 20, 50, 100, 250, 500, 1,000, 2,500, 10,000, 50,000 per group

# Approximation ratios across table sizes

Different-skew margins, common MI 0.02 nats; a ratio of one indicates agreement




*Rows: mean of $\hat{V}/V$; population-input and sample-based variance predictions / observed variance. Thesis Figure 6.1*

# Near-independence noise calculation

Setting: 4×4, n = 1,000, MI 0.0001; $d = 9$ and $\kappa = 2nI(P) \approx 0.2$

$$G = 2n\hat{I}(P) \;\dot{\sim}\; \chi^2_d(\kappa), \qquad \hat{\nu}_P \approx G/2$$

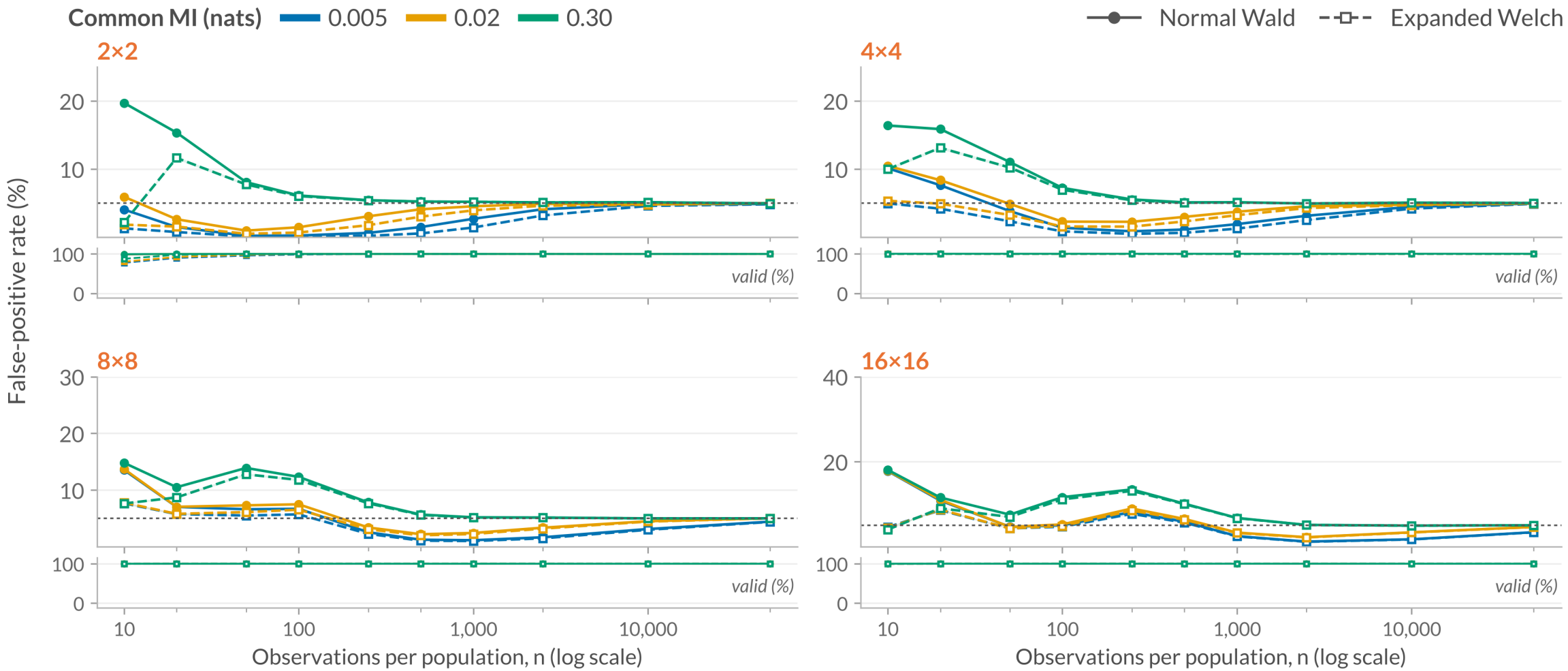
In $G, \hat{I}(P)$ is the uncorrected plug-in MI, not a bias-corrected estimate.

| Quantity | Local prediction | Observed (P) |
|---|---|---|
| Mean estimated component df | 4.60 | 4.86 |
| Sampling-moment target | 9.00 | 9.12 |
| Mean sample-based variance prediction / observed sampling variance | 1.96 | 1.97 |

*Local approximations under fixed dimensions and positive limiting cell probabilities. Thesis Appendix A.2, Section 6.1.3*

# Full null results: uniform margins

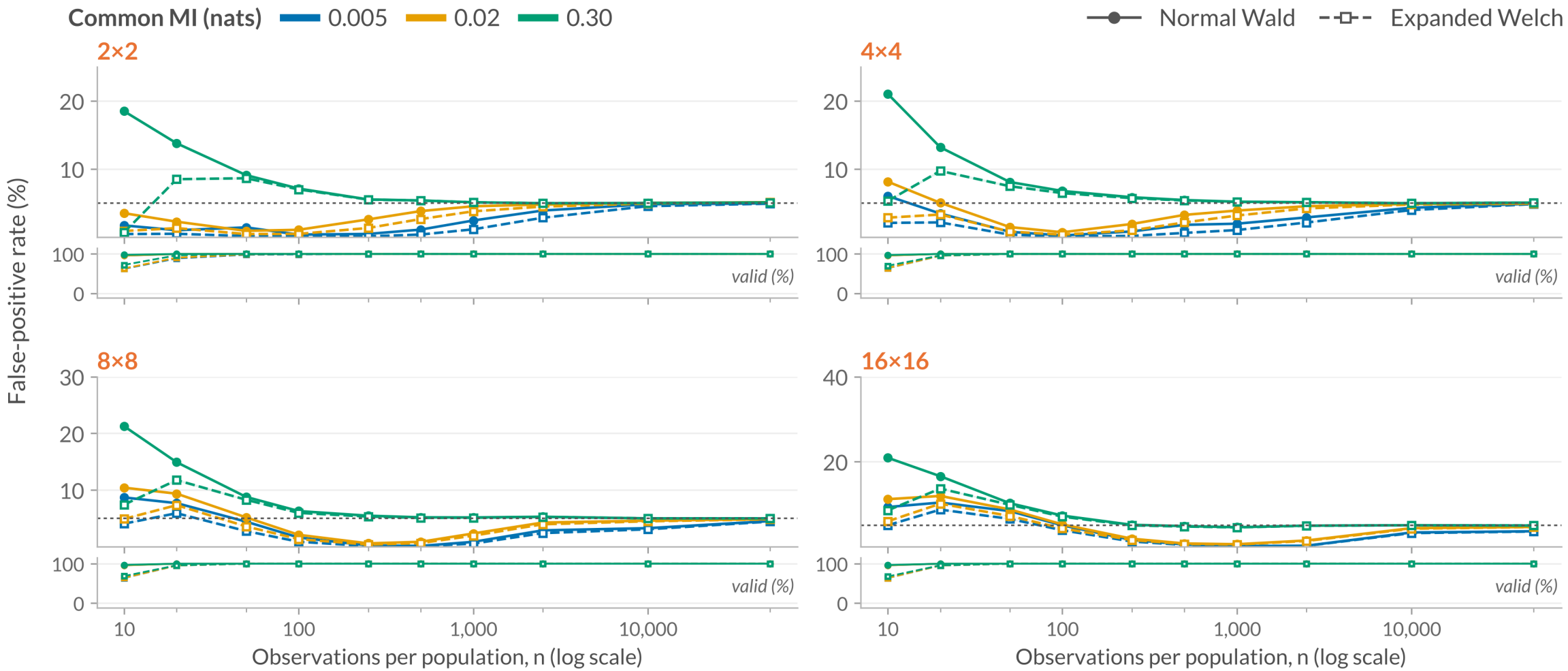
Equal MI in P and Q; uniform margins in both populations




*Rates divide valid rejections by all 100,000 repetitions; invalid outputs count as non-rejections. Rules in B3. Thesis Figure 6.4*

# Full null results: same-skew margins

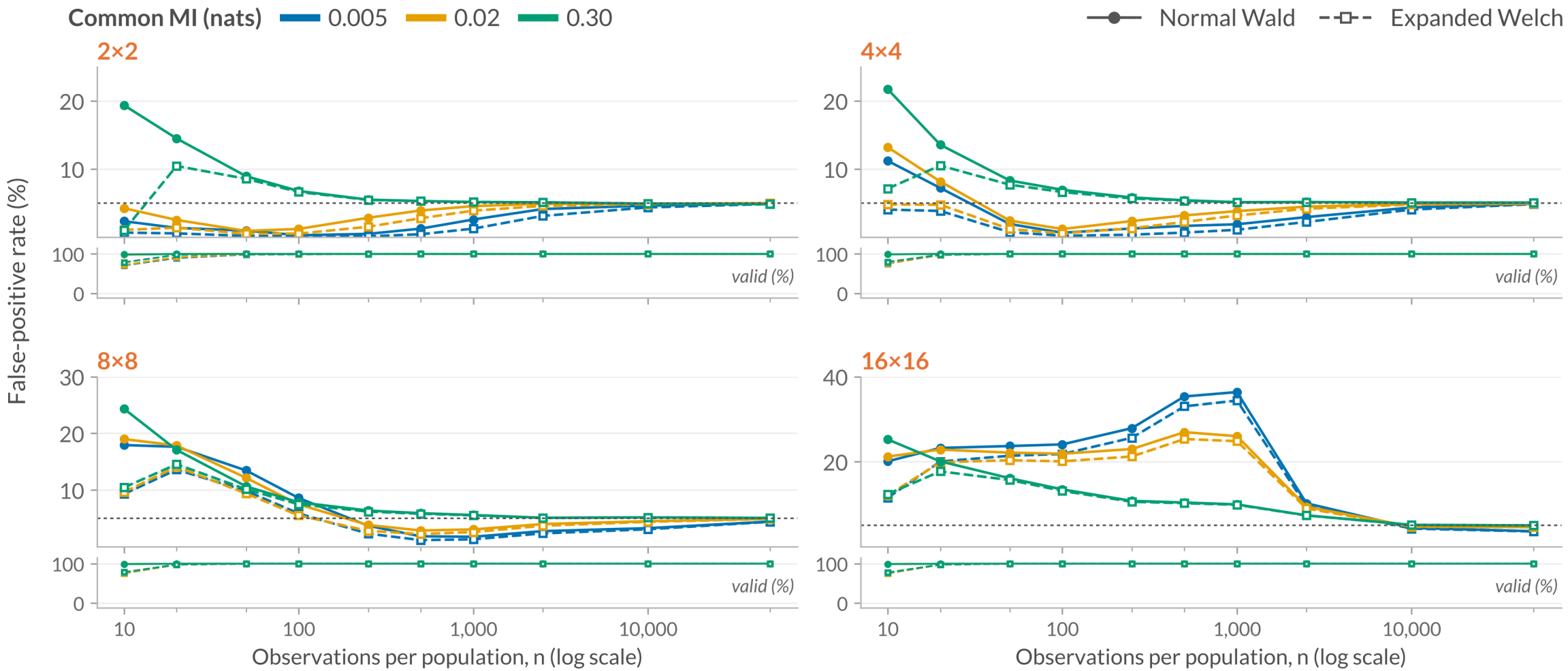
Equal MI; same-skew margins $m_k(0.8)$ in P and Q




*Rates divide valid rejections by all 100,000 repetitions; invalid outputs count as non-rejections. Rules in B3. Thesis Figure 6.5*

# Full null results: different-skew margins

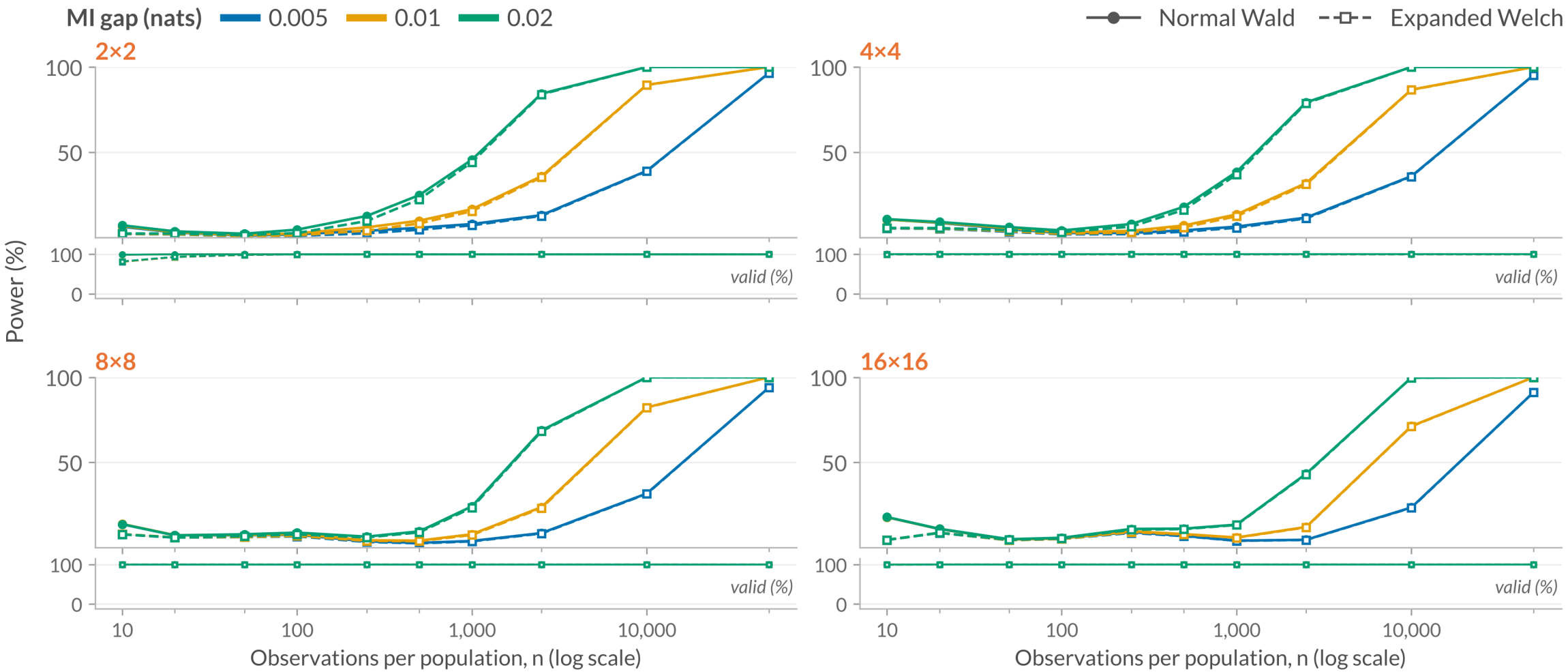
Equal MI; $m_k(0.7)$ in P and $m_k(0.8)$ in Q




*Rates divide valid rejections by all 100,000 repetitions; invalid outputs count as non-rejections. Rules in B3. Thesis Figure 6.6*

# Full power results: uniform margins

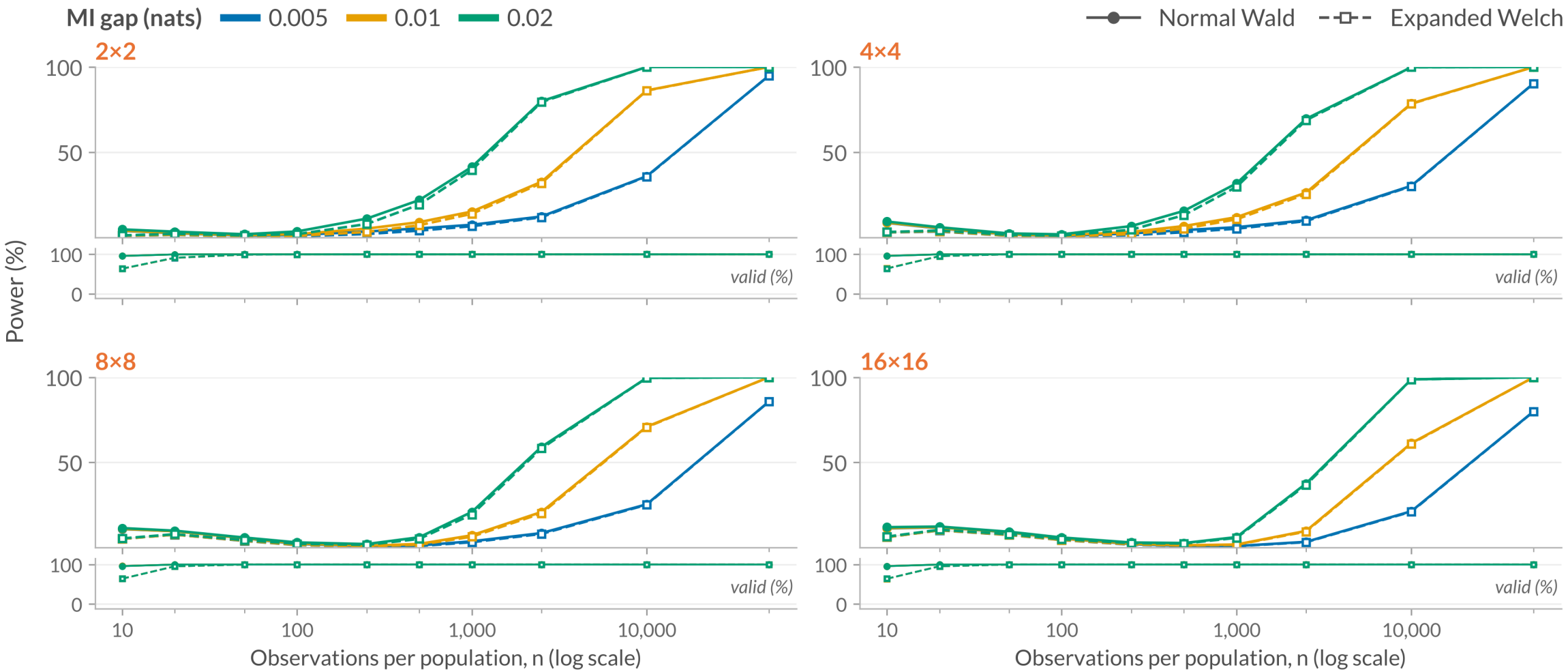
Uniform margins; MI in P = 0.02 nats




*Rates divide valid rejections by all 100,000 repetitions; invalid outputs count as non-rejections. Rules in B3. Thesis Figure 6.7*

# Full power results: same-skew margins

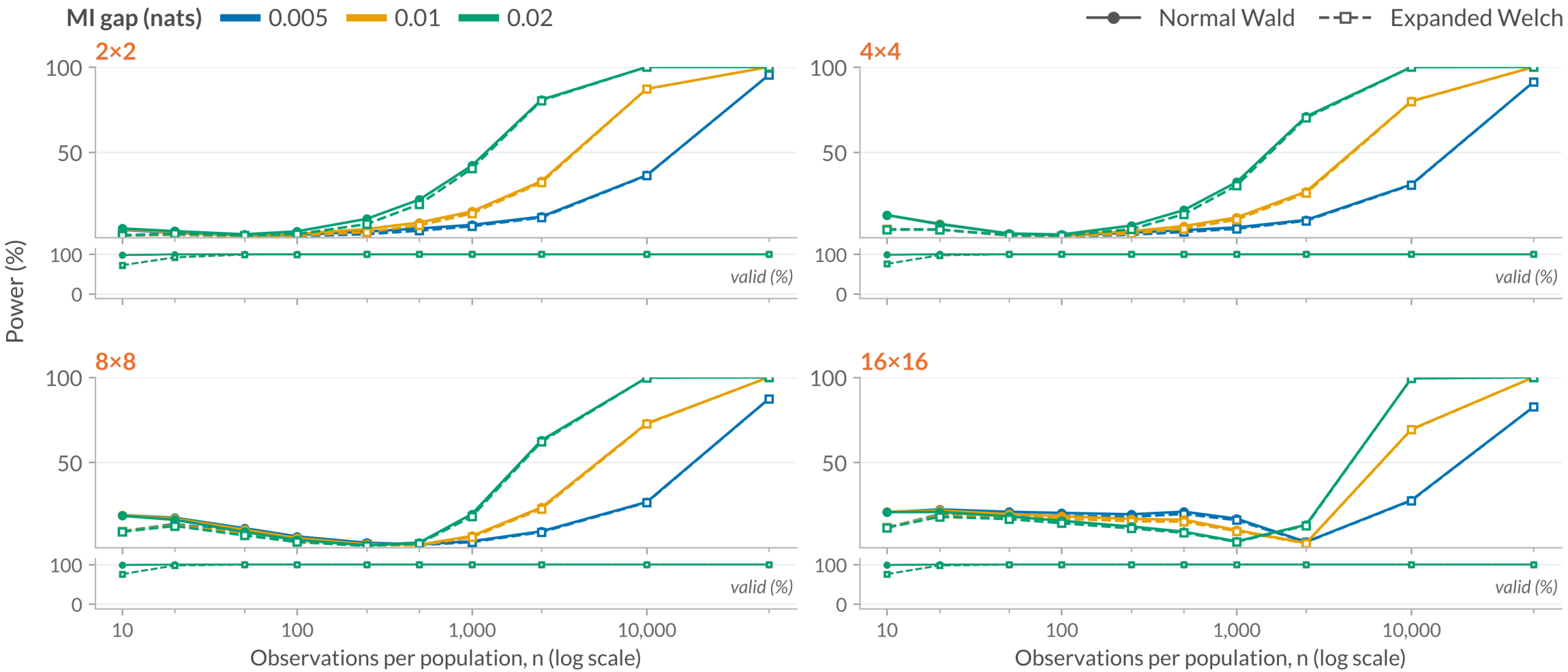
Same-skew margins $m_k(0.8)$ in P and Q; MI in P = 0.02 nats




*Rates divide valid rejections by all 100,000 repetitions; invalid outputs count as non-rejections. Rules in B3. Thesis Figure 6.8*

# Full power results: different-skew margins

$m_k(0.7)$ in P, $m_k(0.8)$ in Q; MI in P = 0.02 nats




*Rates divide valid rejections by all 100,000 repetitions; invalid outputs count as non-rejections. Rules in B3. Thesis Figure 6.9*

# Application details and comparison family

Unadjusted p-values for six exploratory comparisons; Holm adjusts each method separately

| Dataset | Predictor | Normal Wald | Expanded Welch |
|---|---|---|---|
| Adult | Occupation | $1.184\times10^{-44}$ | $1.635\times10^{-42}$ |
| Adult | Education | $1.890\times10^{-34}$ | $4.930\times10^{-33}$ |
| Online Shoppers | Month | $4.714\times10^{-5}$ | $3.726\times10^{-4}$ |
| Student Performance | Study time | 0.1982 | 0.2339 |
| Student Performance | Maternal education | 0.8361 | 0.8468 |
| Online Shoppers | Weekend | 0.9554 | 0.9601 |

Holm-adjusted month p-values: Normal Wald **0.000189**; Expanded Welch **0.00149**

*Becker and Kohavi (1996); Sakar and Kastro (2018); Cortez (2008); Holm (1979). Thesis Appendix B.7, Table B.1*

# References: MI and testing foundations

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